Truck ‘n Trailer problem

MECENG C231A

Final Project

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Overview

01 Problem Statement

02 Dynamics Models

03 Constraints

04 Cost Function

05 MPC Implementation

06 Results

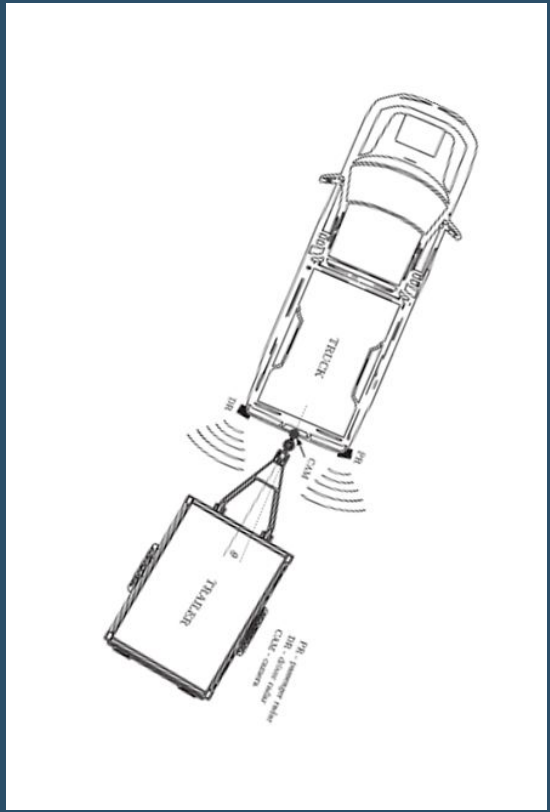
07 Other Findings

08 Conclusion

01 - Problem Statement

Nature of the truck-trailer system:

- Stable in Forward Motion.
- Unstable in Reverse Motion,
- Jackknife Effect: Rapid unstable divergence in hitch Angle



01 – Problem Statement Tasks:

Task 1.0

Forward to a point

Task 1.1

Backward to a point

Task 2.0

Forwards on a path

Task 2.1

Backward on a path

Task 3.0

Single Obstacle (Forward)

Task 3.1

Multiple obstacle (Forward)

Task 3.2

MPC Validation: Randomized Obstacle (Forward)

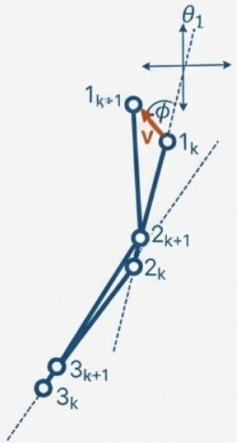
Task 4.0

Backwards Obstacle avoidance

02 – Dynamic Models

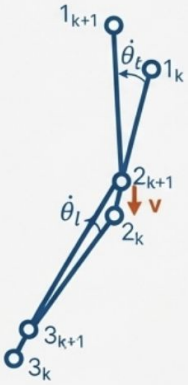
We evaluated three kinematic models. Our goal was realism without excessive computational cost.

Model 1: Geometric



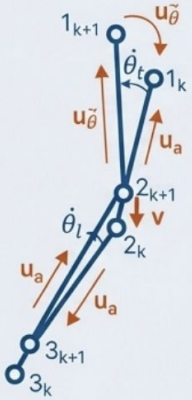
Conceptually simple but less suited for an optimal controller.

Model 2: Bicycle



A standard literature model using state variables. More robust.

Model 3: Improved Bicycle (Chosen)



Our final choice. By using acceleration and steering rate as inputs, it ensures smoother, more physically realistic vehicle motion.

02 – Dynamic Models

Forward Euler Discretization

6 State Variables

2 Inputs

$$q(k+1) = q(k) + \Delta t \cdot \dot{q}(k).$$

$$\dot{x}_2 = v \cdot cos(\theta_t)$$

$$\dot{y}_2 = v \cdot sin(\theta_t)$$

$$\dot{\theta}_t = \frac{v}{L} \cdot tan(\phi)$$

$$\dot{\theta}_l = \frac{v}{d} \cdot tan(\theta_t - \theta_l)$$

$$\dot{v} = a$$

$$\dot{\phi}_{State\ Variable\ Derivative} = \dot{\phi}_{Input}$$

03 – Constraints

Core Model:

- Velocity
- Steering Angles

- Steering Rate
- Acceleration

- Jackknife Angle.

$$v_{min} \leq v_k \leq v_{max}$$

$$\phi_{min} \leq \phi_k \leq \phi_{max}$$

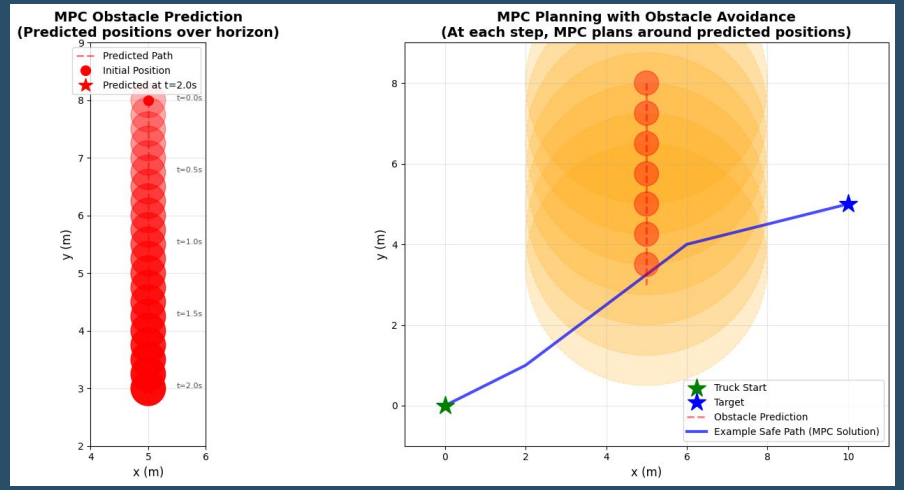
$$a_{min} \leq a_k \leq a_{max}$$

$$\dot{\phi}_{min} \leq \dot{\phi}_k \leq \dot{\phi}_{max}$$

$$-\psi_{max} \leq \theta_{t,k} - \theta_{l,k} \leq \psi_{max}$$

03 – Constraints

Obstacle Avoidance:



Truck n' Trailer

$$(x(t) - x_o(t))^2 + (y(t) - y_o(t))^2 \geq r_{safe}^2$$

Obstacle

Obstacle Prediction

$$p_o(t) = p_o(o) + v_0 t$$

04 – Cost Function

The cost function minimizes:

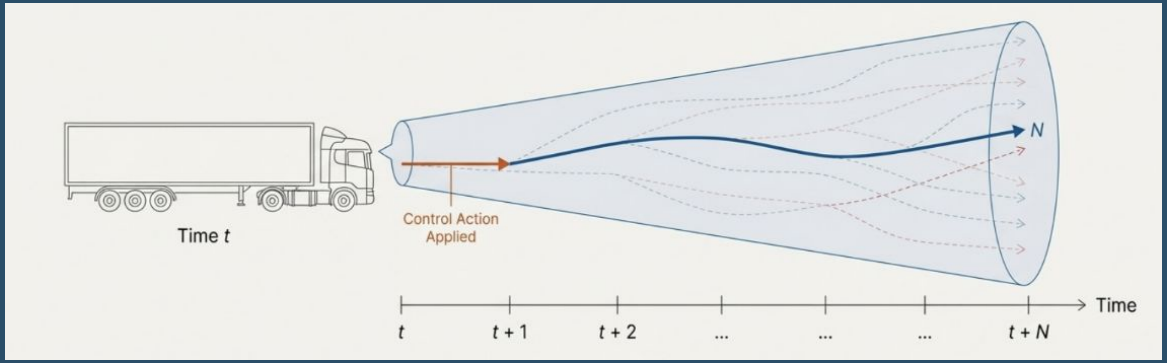
- Difference between current states & final states
- Velocity and angular velocity

$$J = \sum_{i=1}^{4} (q_i(N) - t_i)^2 + W_i \cdot \sum_{k \text{ in } K} (a_k^2 + \dot{\phi}_k^2)$$

05 – MPC Implementation + Warm Start Strategy

MPC workflow:

- Measure current state
- Solve CFTOC
- Implement the first input
- Shift horizon and repeat



05 – MPC Implementation + Warm Start Strategy

Warm start strategy:

- Use previous prediction as Initial guess

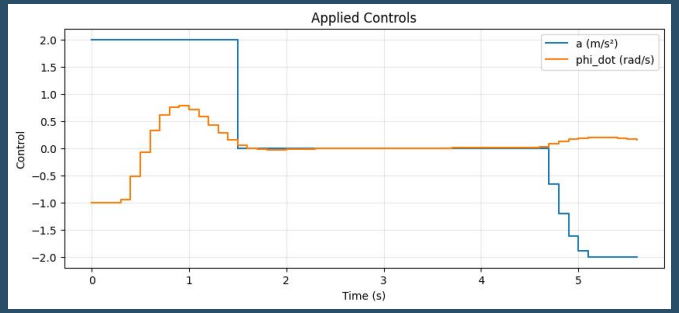
Advantages

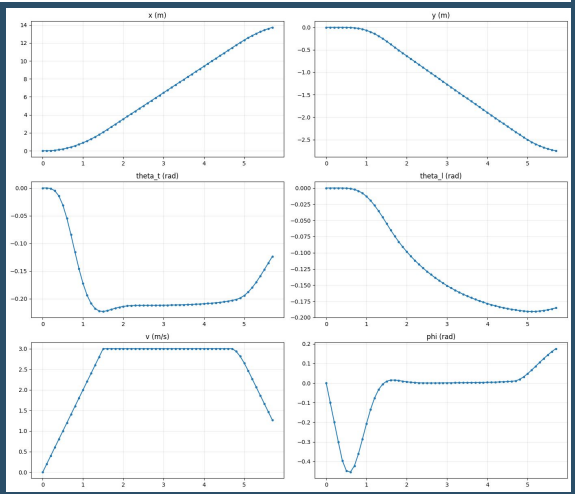
- Less iterations
- Convergence

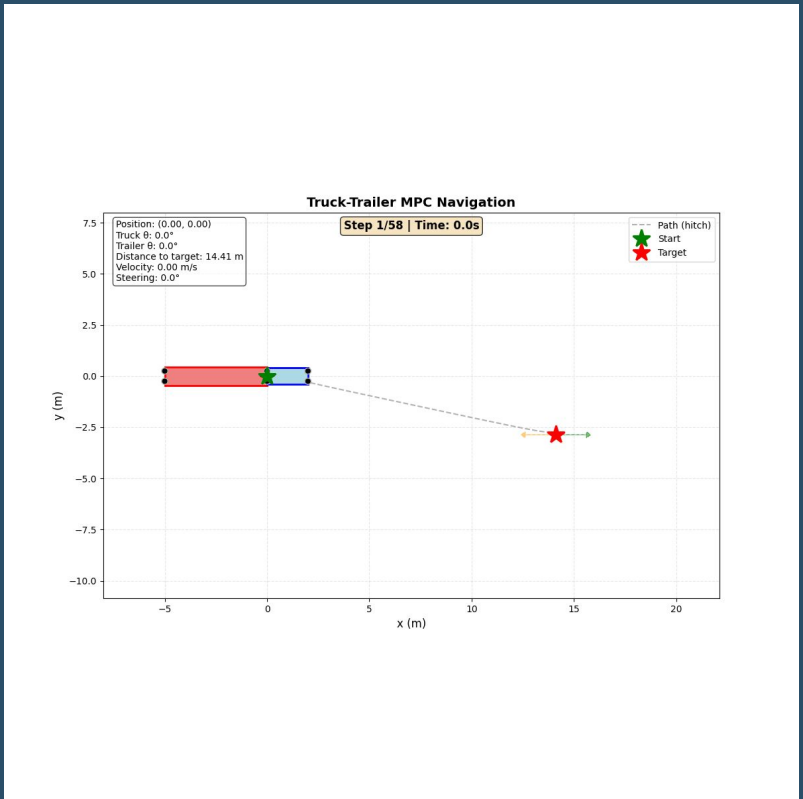
$$u^{guess}_{0:N-2} \leftarrow u^{\star}_{1:N-1} \qquad u^{guess}_{N-1} \leftarrow u^{\star}_{N-1}$$

$$x^{guess}_{0:N-1} \leftarrow x^{\star}_{1:N-1} \qquad x^{guess}_{N} \leftarrow x^{\star}_{N}$$

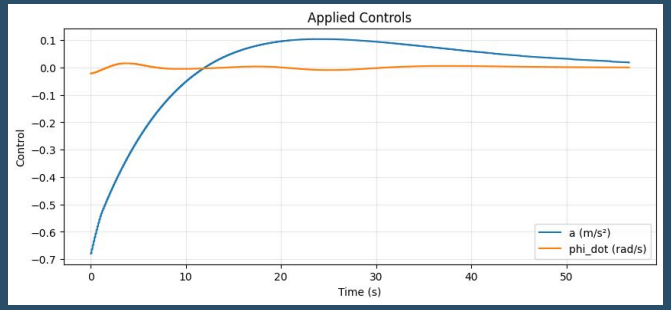
06 – Results – First Dynamics Testing

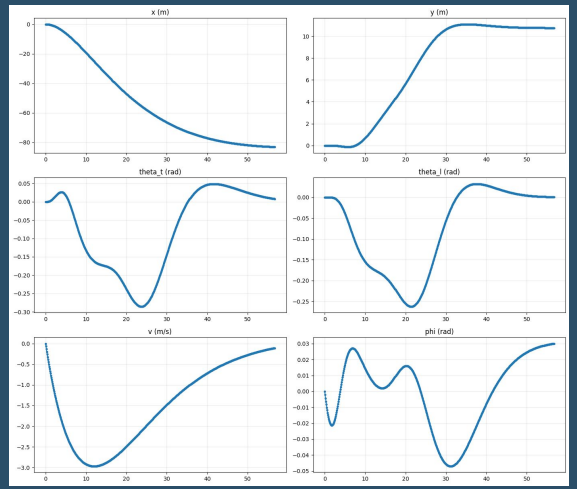


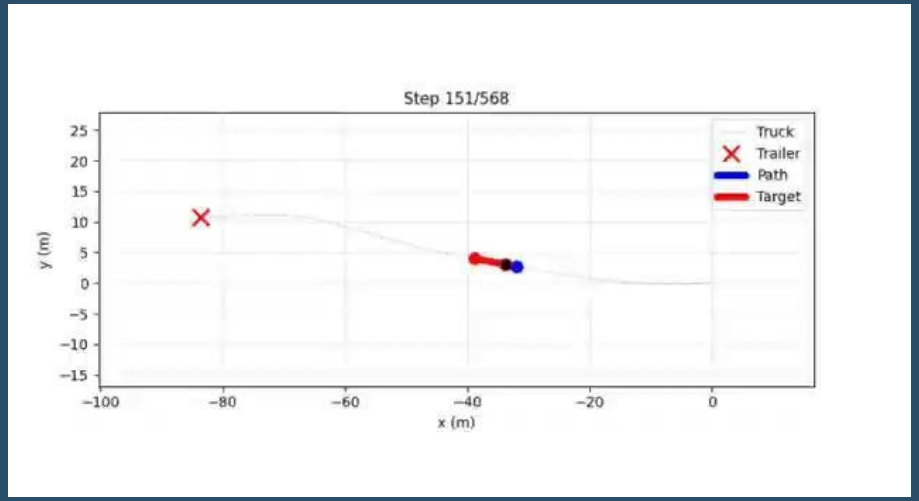




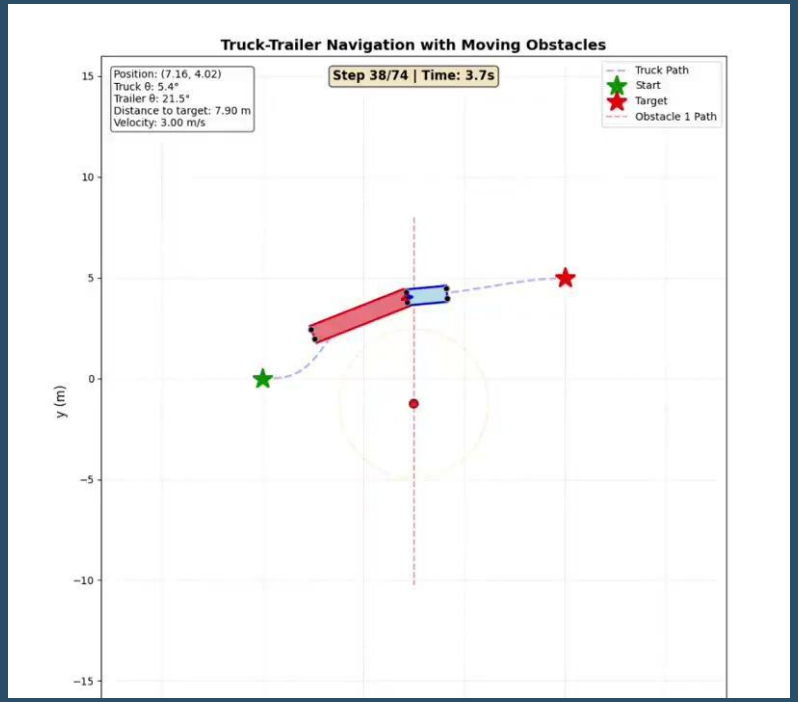
06 – Results – The First Triumph: The Reverse Maneuver

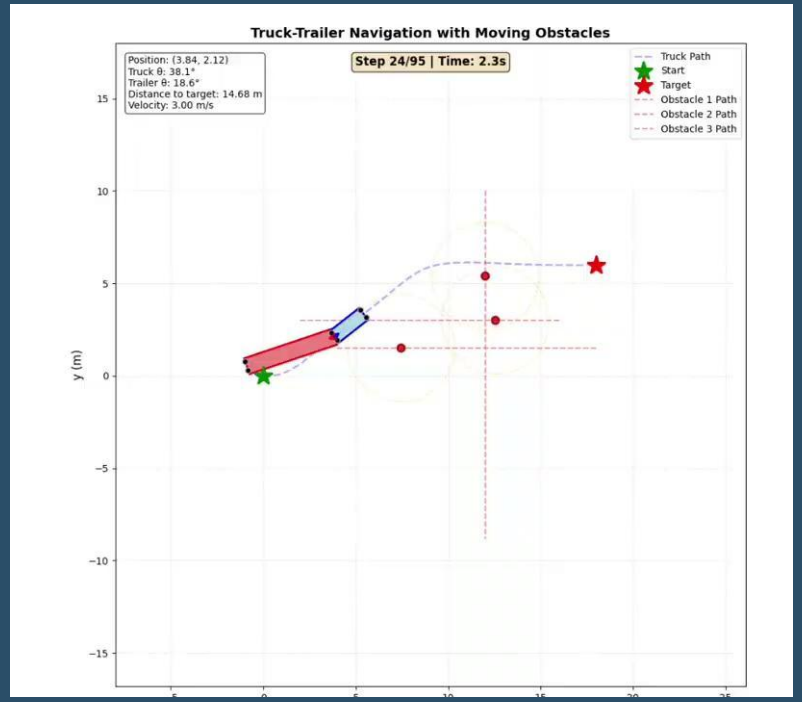




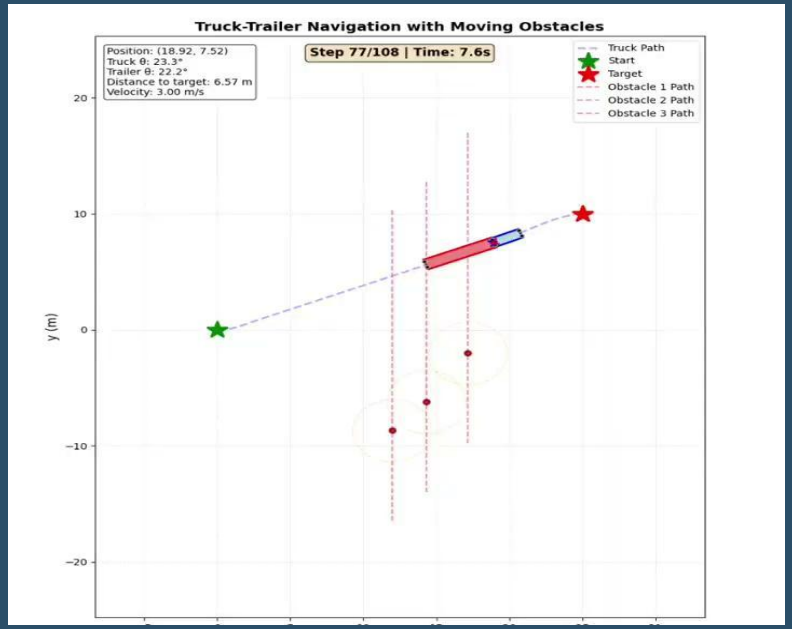


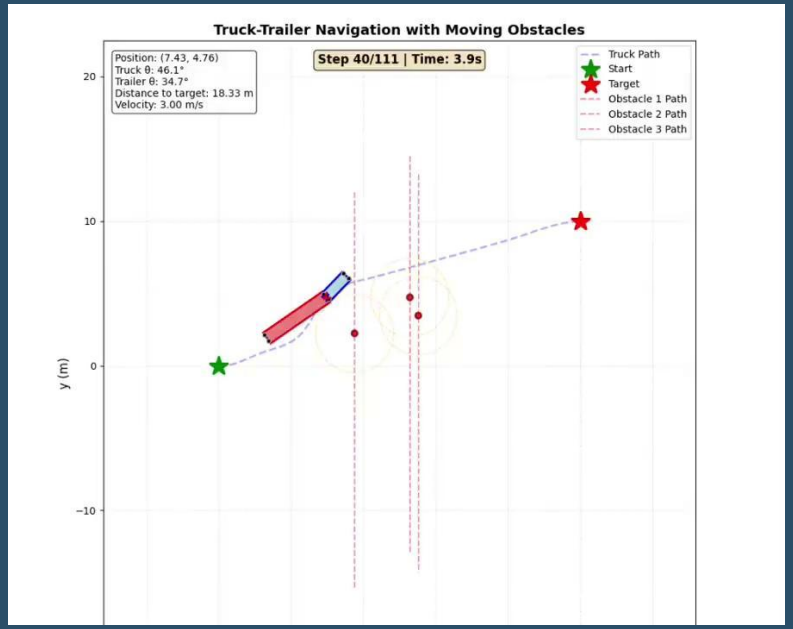
06 – Results – Enforcing Obstacles Constraints





06 – Results – Comparison of our MPC Model

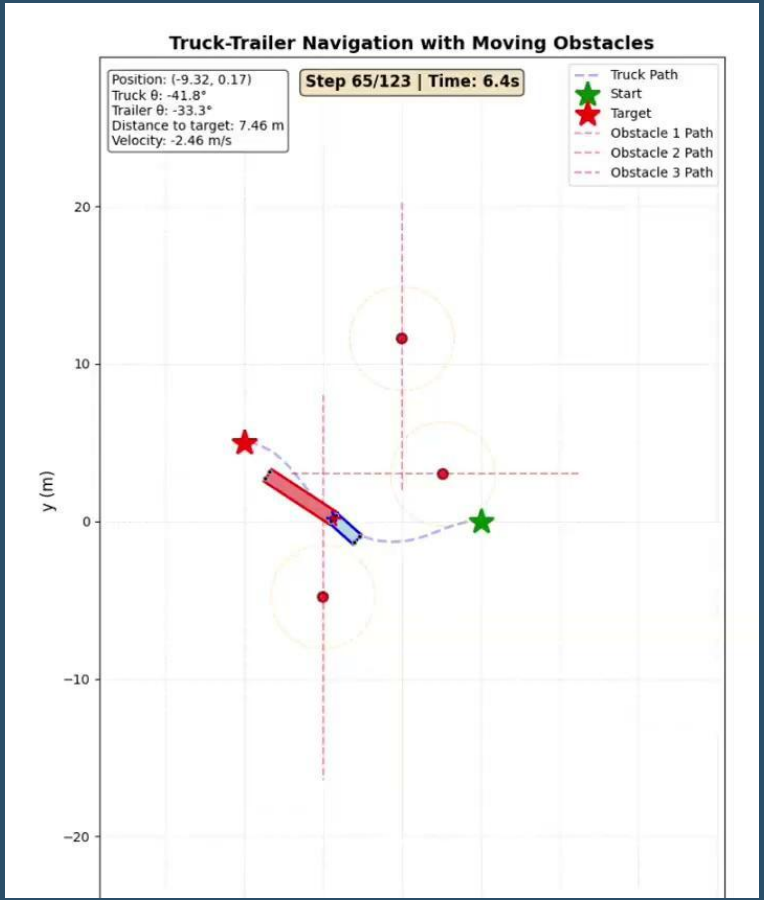




06 – Results

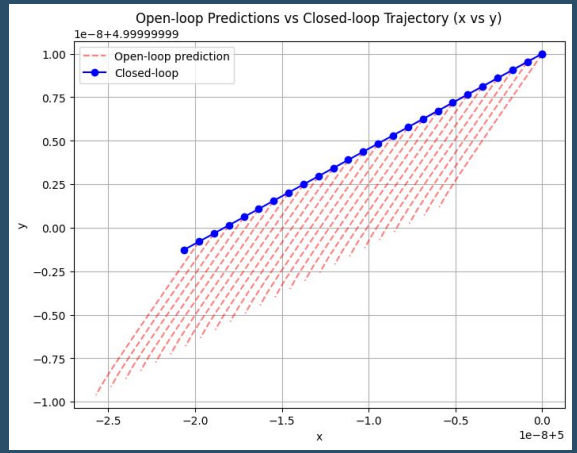
The Ultimate Test: Navigating a Dynamic World

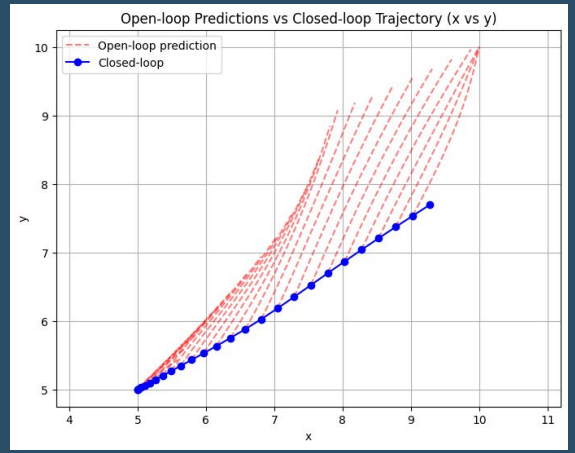
The final system combines path tracking and predictive obstacle avoidance. The MPC plans a safe trajectory around moving object by treating collision avoidance as a non-negotiable constraint.

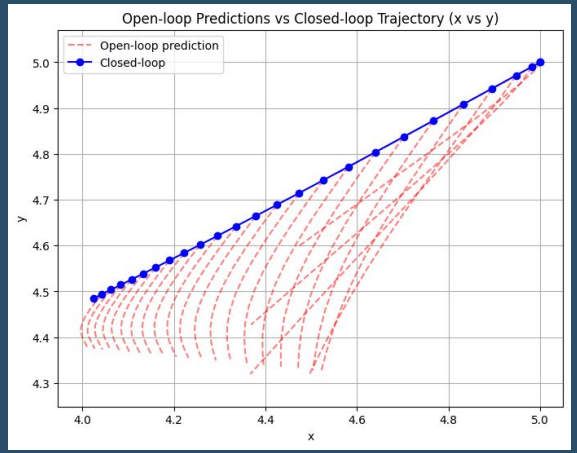


07 - Other Findings

Alternative Approach : Linearized dynamics and P-Q-R cost matrices







08 – Conclusion

What we got

1. Stabilized Reversing
2. Safe pathfinding through obstacles

What could be improved

1. Dynamic modelling beyond bicycle
2. More efficient code for real-time-compute

THANK YOU